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Editor-in-Chief
IEEE Access

Dear Editor-in-Chief,

We would like to submit the enclosed manuscript entitled “*DC-Link Voltage Scaling to 1.2 kV for Heavy-Duty EVs: System-Level Assessment with Experimental Validation of SiC Traction Inverter*” for consideration for publication in *IEEE Access*.

This paper presents a system-level investigation of increasing the dc-link voltage from 800 V to 1200 V in heavy-duty electric vehicle (HDV) powertrains, considering its implications for charging infrastructure, traction inverter design, and motor harmonic losses. A virtual-prototyping framework is developed to evaluate inverter losses, switching-frequency optimization, dc-link capacitor requirements, and power density under practical dv/dt and thermal constraints. The proposed methodology is further validated experimentally using a 250 kW, 1.2 kV SiC-based traction inverter prototype. The analytical predictions show good agreement with the experimental results across multiple operating points, with deviations below 10%.

In contrast to prior studies that primarily focus on subsystem-level analysis or passenger electric vehicles, this work targets HDV-scale powertrains and explicitly examines the impact of higher dc-link voltage on both efficiency and physical design metrics. The results show that, under representative practical constraints, increasing the dc-link voltage to 1200 V reduces charging losses, decreases dc-link capacitor volume, and improves inverter power density without increasing PWM-induced motor harmonic losses.

The manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have read and approved the manuscript and agree with its submission to *IEEE Access*.

We believe that the broad scope, system-level perspective, and experimentally validated results make this work well suited to *IEEE Access* and relevant to researchers and practitioners in power electronics, electric drives, and electrified transportation systems.

Sincerely,

Enes Ayaz
On behalf of all authors